Exam 1 Review 1 DE

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1 Introductory Remarks

These reviews are designed to reflect both the format and difficulty of Professor Ambrose's exams. I encourage you to use them as a guide for identifying the material you should focus on when preparing for each exam.

Differential Equations is a difficult class and requires cumulative preparation. Please note that these packets are only meant to supplement your studying. They do not replace the valuable learning that comes from attending lectures or working on homework.

Solution manuals are provided for each review.

2 Problems

2.1 Problem 1

Solve the DE

$$\frac{dy}{dx} = \frac{xe^{x^2}}{y(y + \ln(y))}$$

2.2 Problem 2

Solve the DE

$$\frac{dy}{dx} = \frac{2y^4 + x^4}{xy^3}$$

2.3 Problem 3

Solve the DE

$$(2x^2y - 2x^3)dy + (4x^3 - 6x^2y + 2xy^2)dx = 0$$

2.4 Problem 4

$$\frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{x}{1+x^2}$$

2.5 Problem 5

A 200-gallon tank initially contains 80 gallons of pure water. Starting at $t = 0$, salt water containing 2.0 pounds of slater per gallon falls into the the tank at5 gal/min. The thoroughly mixed solution flows out at 3 gal/min. How much salt is in the tank when the tank overflows?

2.6 Problem 6

At $t = 0$, a 48-pound object is pushed beneath the surface of a calm lake and then released from rest. The object has a buoyancy force of 57 pounds, so it begins to rise after release. The resistance of the water (in pounds) is given by $\frac{v^2}{4}$, where v is the upward velocity of the object in feet per second. If the object reaches the surface in 3 seconds, what is the velocity of the object when it reaches the surface of the lake?

2.7 Problem 7

He combined weight of a boy and his sled is 120 pounds. Starting from rest at $t = 0$, he sleds down a hill whose slope is $\frac{3}{4}$. The air resistance (in pounds) is numerically equal to twice the velocity (in feet/sec) of the boy and his sled. If the coefficient of sliding friction between the sled's runners and the sloping hill is $\frac{1}{12}$, (a) what is the velocity of the sled at the end of t seconds, and (b) how long (correct to the nearest foot) is the hill if it take 6 seconds to reach the bottom of the hill?